mNGS

Study design and sample prepration

Key points to learn in this session

Good study is essential for mNGS an requires an understanding of:

- Purpose of the study (e.g. clinical diagnostics, outbreak detection)
- Pathogen landscape
- Sample type
- Controls
- Sample selection and metadata
    - Sensitivity of mNGS (*e.g. are we getting enough reads?*)

Study design



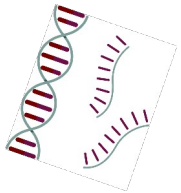
mNGS pipeline

Study design



- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction



- Type(s)
- Controls
- QC

Library prep



- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

Sequencing



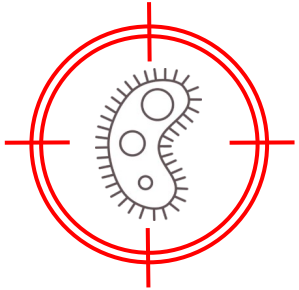
- Machine
- Read length
- Loading conc.

Data analysis



- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

Pathogen ID



Good study design is essential for mNGS!



It requires an definition of:

- **Purpose of the study (e.g. clinical diagnostics, outbreak detection)**



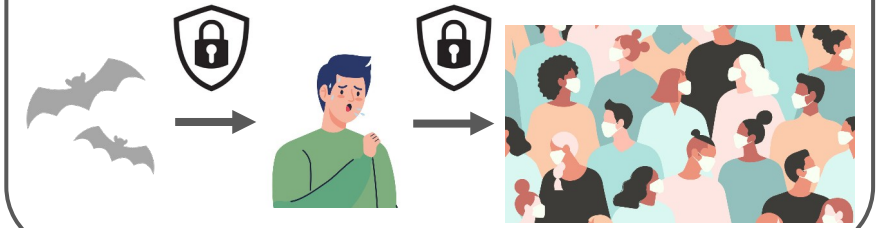


Some use cases for mNGS

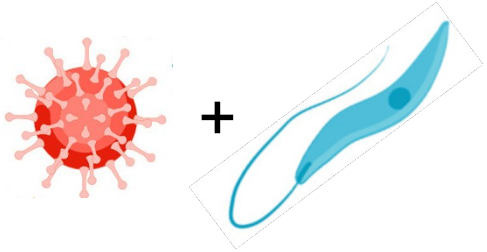
Identify a novel pathogen



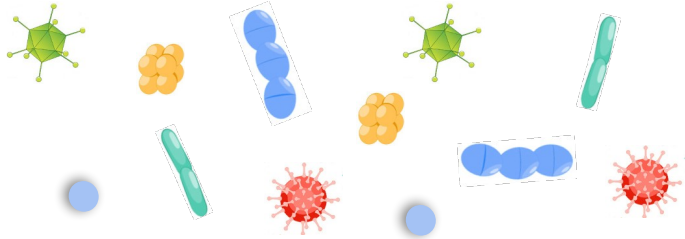
Outbreak detection and prevention



Identify coinfections



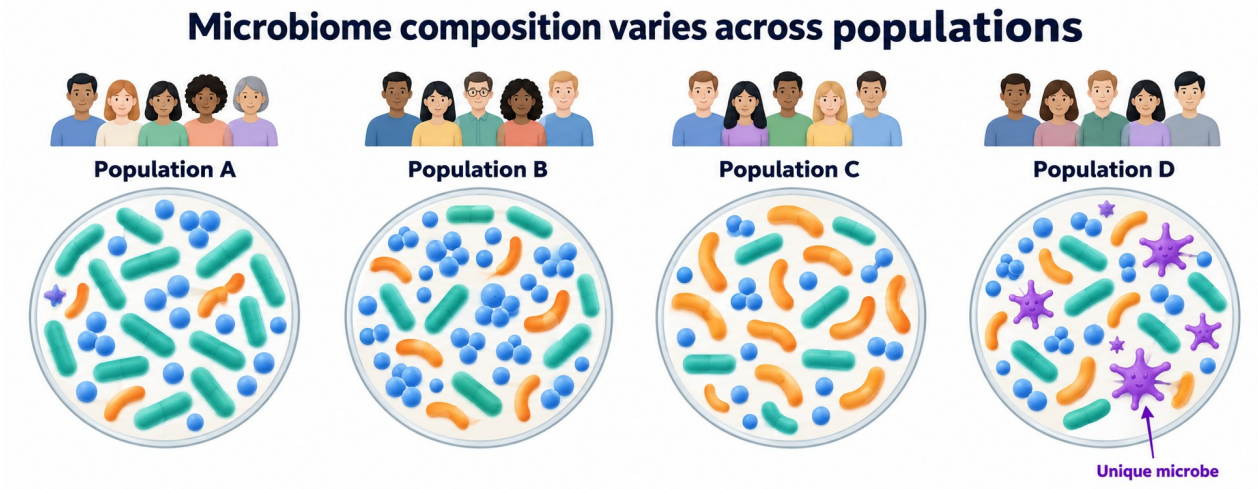
Characterize the microbial landscape



Pathogen landscape



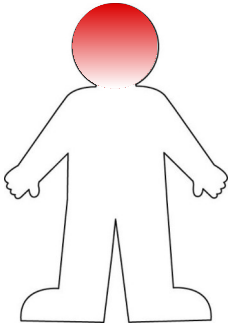
What pathogens are suspected, what other microorganisms are we expecting



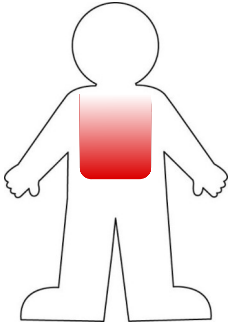


Sample type

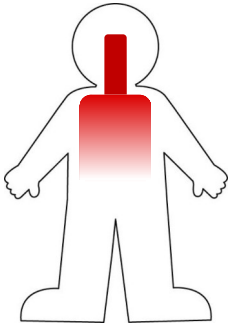
How to choose the sample type? Depends on the question!

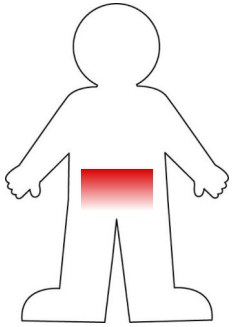


Neurological symptoms?
Meningitis/Encephalitis?
CSF and Serum



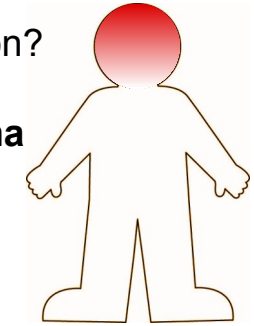
Respiratory symptoms?
Upper = **NP, OP swab, saliva**
Lower= **Bronchial lavage, pleural fluid, lung biopsy**



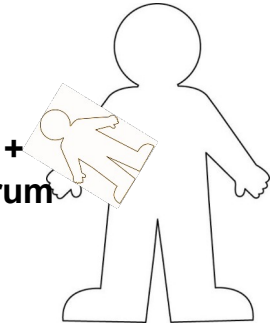


Gastrointestinal symptoms?
Stool, urine, rectal swab, vaginal swab

Systemic infection?
Rash?
Serum or plasma



Neonatal infection?
Paired sample of neonatal CSF/Serum + rectal, vaginal, or serum sample from mother



Different human sample types have different RNA contents!

Cerebral Spinal Fluid
No infection: 0% microbial
With infection: <1% - 5%

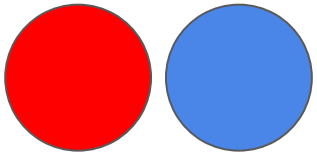
Nasal or oral swab
50% microbial (mostly bacteria)

Whole Blood: lot of background
No infection:
During infection: <<<0% microbial

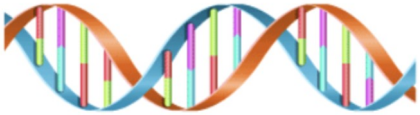
Tracheal Aspirate
5% microbial, 95% human

Serum or Plasma
10-100x less host material

Rectal or Vaginal swabs or stool
50% microbial
(mostly bacteria)

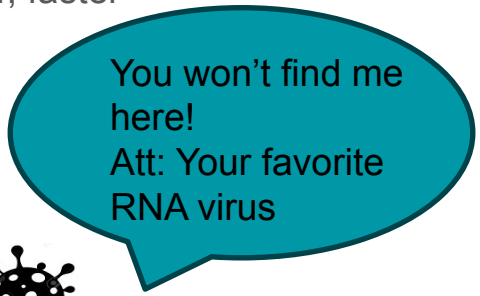


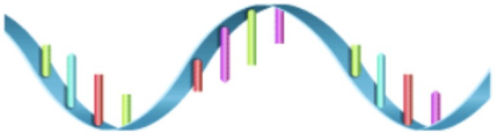
Type of nucleic acid: DNA Vs RNA



Everything that's there
…Even if it's dead

- Well suited for pathogens with 'large' genomes
- Can have too much host signal
- Easier, faster





Everything that's **Alive**
…Including *that* RNA virus

- Most will be rRNA
- RNA is labile
- Excellent preservation is mandatory

In some cases, there are depletion methods available



Study controls are CRUCIAL to make sense of sequencing data

- Healthy controls: clinical samples from healthy controls
  - What are organisms that are commonly found in your population?

- Sampling: mock samples (e.g. needle touching but not piercing skin during CSF sampling)
  - What are the common contaminants that occur during the sampling process?

- During nucleic acid extraction: water controls
  - What are the common contaminants that occur during extraction?
  - What is the level of cross-contamination that occurs between samples during extraction?

Study subject include known positive and negative samples



Metadata

Diagnostic test results

WBC PCR Ab/Ag Cultures

Dates

Symptoms Location

Age

+

Study subjects



Unknown cases of interest

Fever with unknown cause

Positive controls



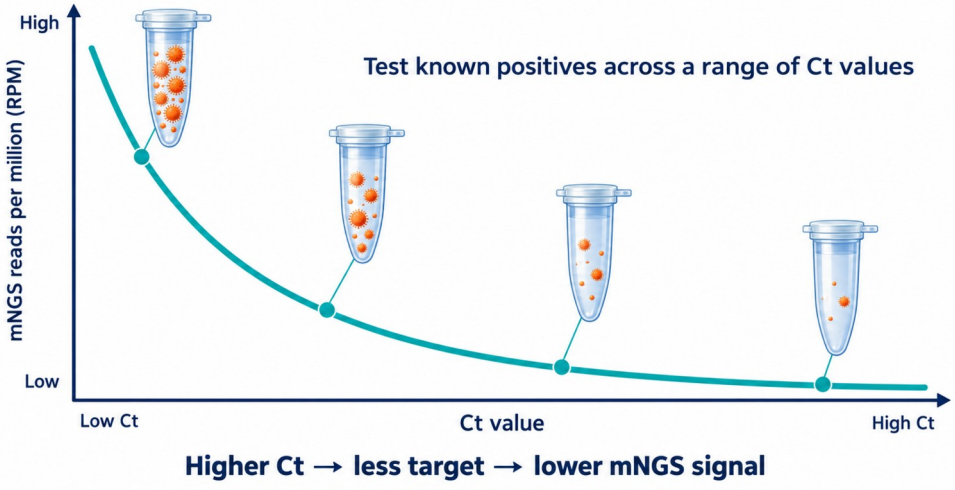
Fever with known cause



Healthy individuals

Negative controls and potential incidental findings

Value of positive samples to assess detection and sensibility



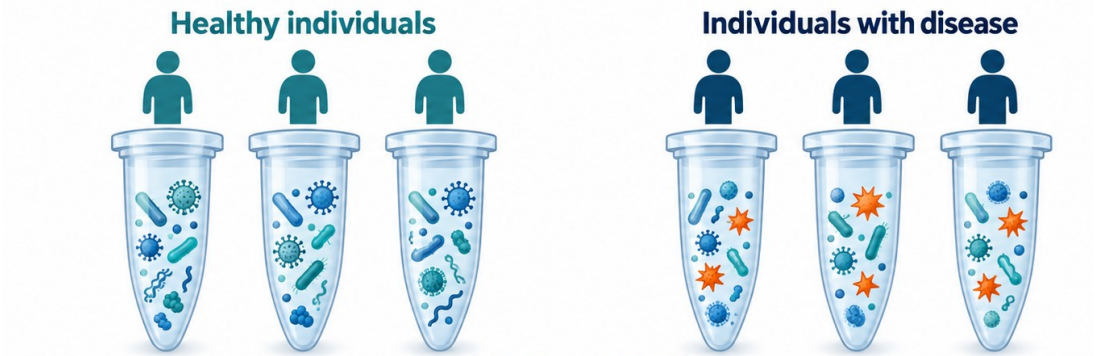
What if you don't find a pathogen?

Did the experiment fail?

Positive controls are the answer, and a range of pathogen types and abundances will give you a sense of sensibility

Ideally the detection method used is dependent on the presence of nucleic acids (qPCR, etc.)

Value of negative samples to assess cross contamination, microbiological landscape and background signal



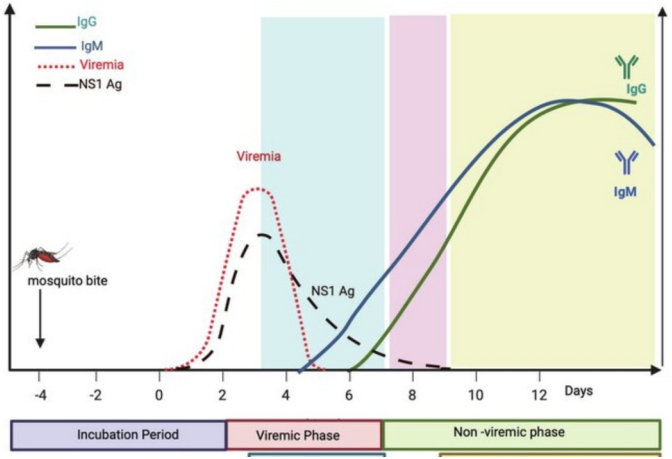
- Samples from ”healthy individuals” help distinguishing background microbiota from true disease-causing agents
- However association alone does not establish causation



Sample selection for sequencing

When studies are retrospective consider prioritizing samples that:

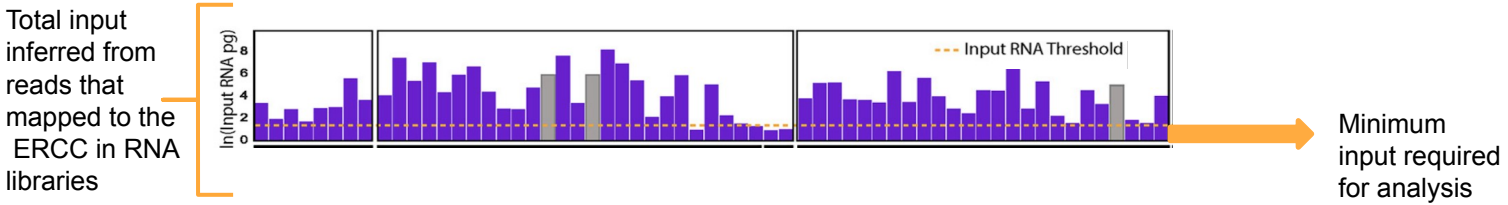
- Were collected during the acute phase of symptom onset
- Presented severe and/or various symptoms
- Had evidence for infectious etiology
- Were well preserved
- Have ample metadata
- Prioritize positive controls tested with NA



DOI:10.5772/intechopen.1006372

Understanding the sensitivity/specificity of mNGS

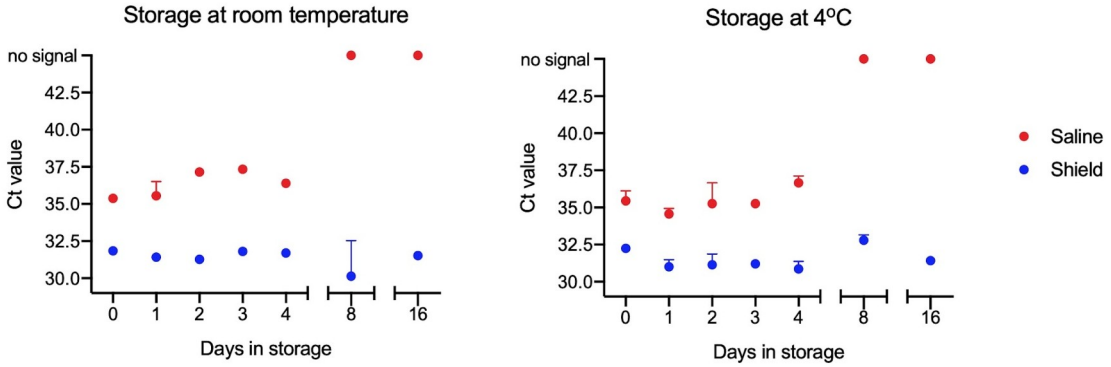
- Sensitivity and specificity depend on:
  - Amount and **quality** of sample input
  - Efficiency of each step in mNGS
  - Accuracy of “base calling” during sequencing
  - Quality of genome database





Sample preservation

- Sample collection:
  - Collect in nucleic acid stabilizers to preserve sample quality
    - e.g. DNA/RNA Shield (Zymo), RNA*later* (Qiagen)
  - Prepare your own collection kits with Shield (cheaper than buying pre-made kits)
  - Collect samples as sterile as possible
  - Collect “collection controls”: e.g. needle touch without aspiration, ...



In vitro transcribed RNA fragments preserve better in Shield than in Saline when stored at room temperature or at 4°C, as assessed by qPCR.



DNA/RNA Shield

- Preserves RNA for 24 hrs at RT, indefinitely at -20°C
- Inactivates pathogens
- 1x Solution for solids & swabs
- 2x Solution for liquids
- -> Order 2x and dilute ½ for swabs if needed



Sample storahe

- Cold chain preserves sample quality



Collect in
DNA/RNA Shield
Or RNA*later*



Keep cold during
transport



Long term storage
at -80°C



Study design

Key points:

✓ Preserve sample quality

✓ Collect metadata

✓ Include controls !

✓ Minimize cross-contamination

- Study subjects
  - Population of interest + metadata
  - Healthy controls (!)
- Sample types:
  - Swabs: oral, rectal, vaginal, ...
  - Blood vs serum
  - CSF
- Sample collection:
  - Collect in nucleic acid stabilizers: e.g. DNA/RNA Shield (Zymo), RNA*later* (Qiagen)
  - Work as sterile as possible and collect appropriate controls
- Sample storage:
  - Keep cold during transport (!)
  - Flash freeze as soon as possible (!)

Example dataset - Fever with unknown etiology

		Known positives						**Unknowns**						**Known negatives**						**Water**	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	W1	W2
Traditional diagnostics	Malaria blood smear results	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Human African Trypanosomiasis serology	-	-	-	+	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-

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mNGS results	*Plasmodium falciparum*																				
	Trypanosoma brucei gambiense																				
	Nipah henipavirus																				
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An adequate set of controls will help you discriminate true pathogens from:

- Contaminants during library preparation (extraction to sequencing library)
- Contaminants from sampling procedure

Background signal

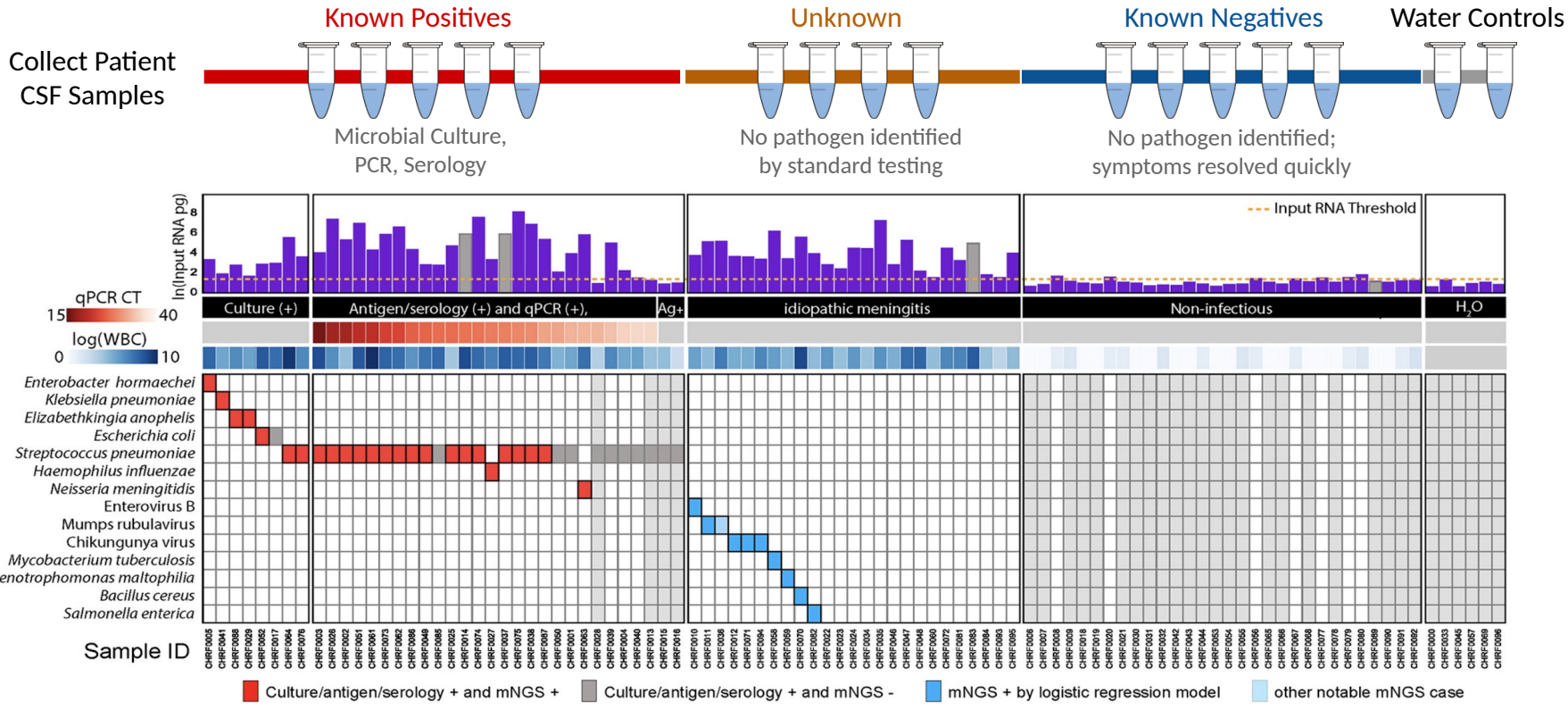
Practice scenario: You have been tasked to conduct a study to investigate a potential outbreak at a wet market. Various types of birds (chickens, ducks) are handled and sold here. Recently, a number of animals, as well as people handling them, have been getting sick with respiratory symptoms.

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1. *What does your suspected pathogen landscape look like?*
2. *What type(s) of samples would you want to collect?*
3. *What metadata would you want?*
4. *What controls would you want to include?*

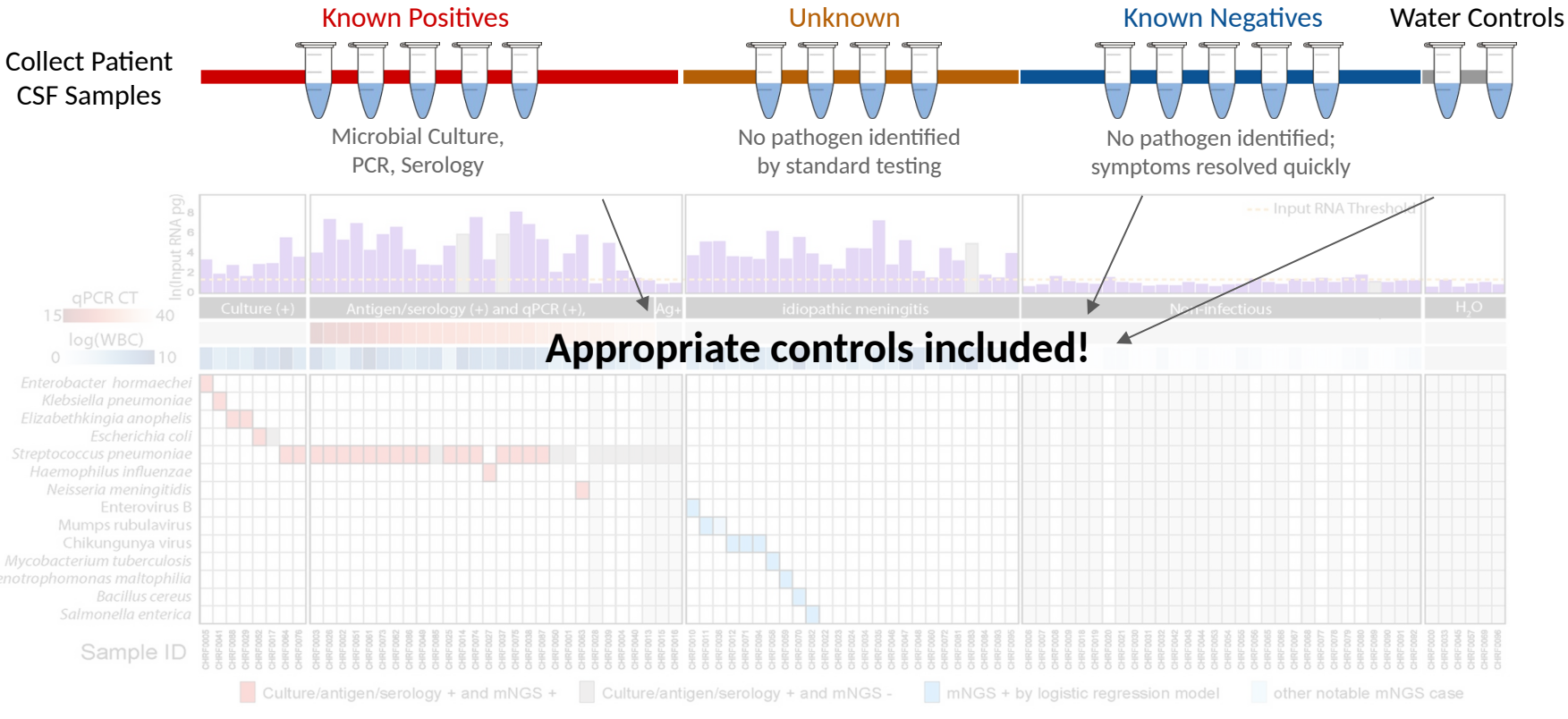
Use case: Saha et al., mBio, 2019

mNGS to identify etiologies of pediatric meningitis in Bangladesh



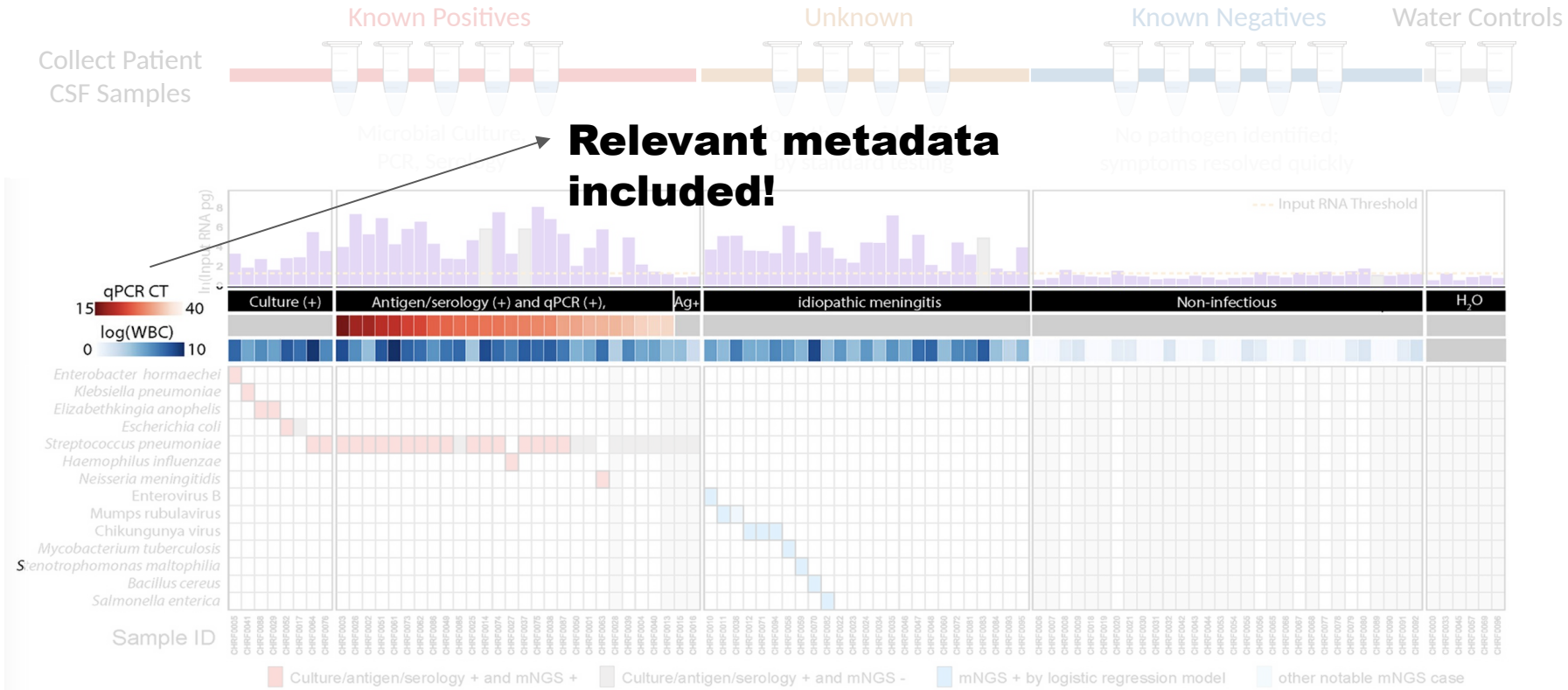
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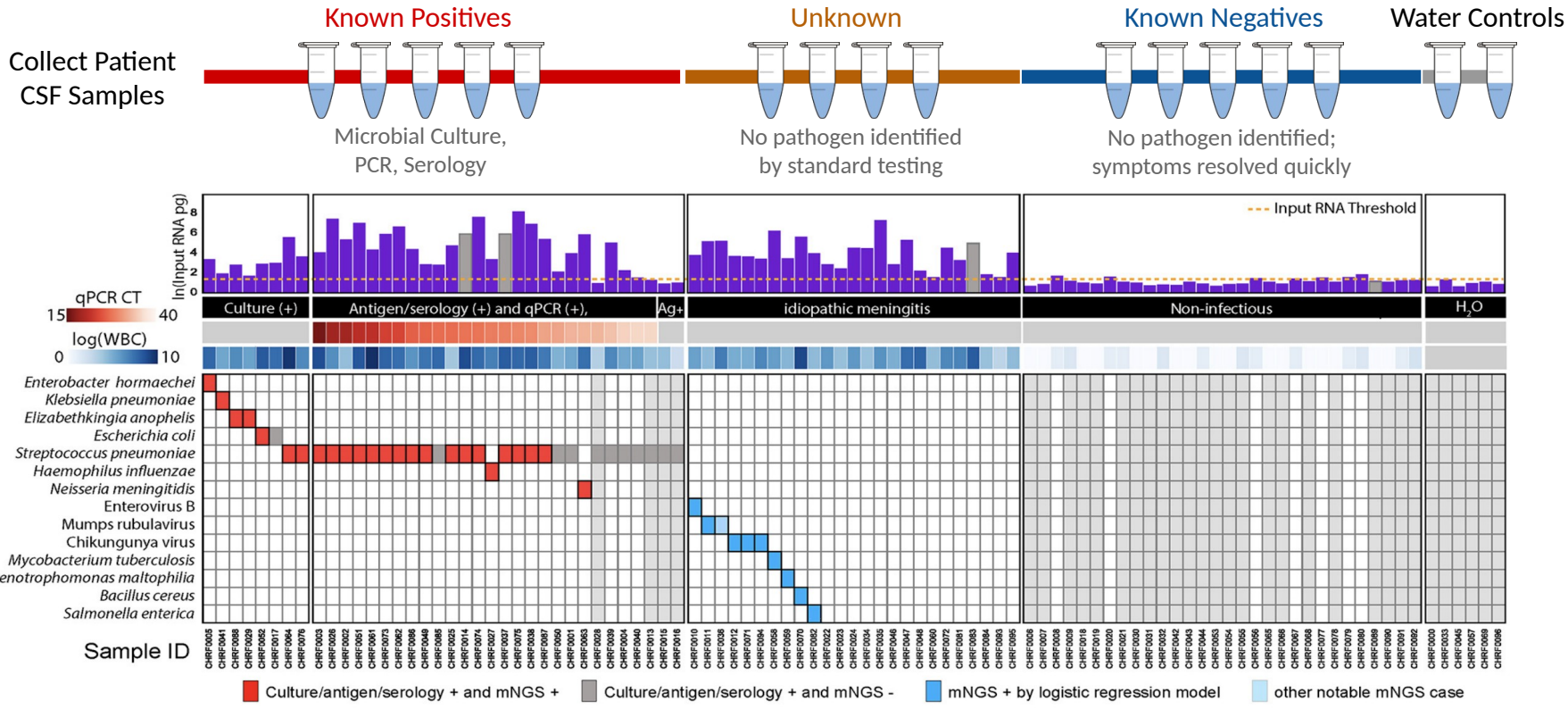
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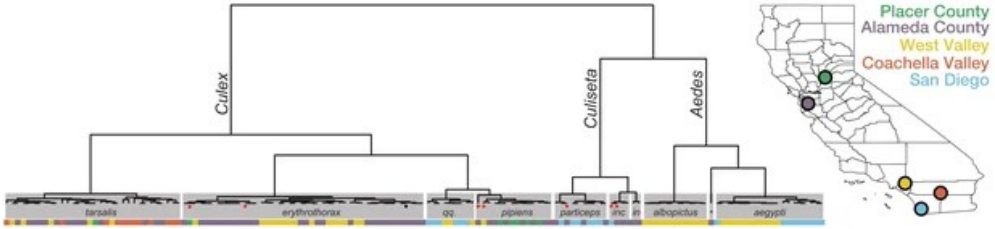


Use case: Batson *et al.*, eLife 2021

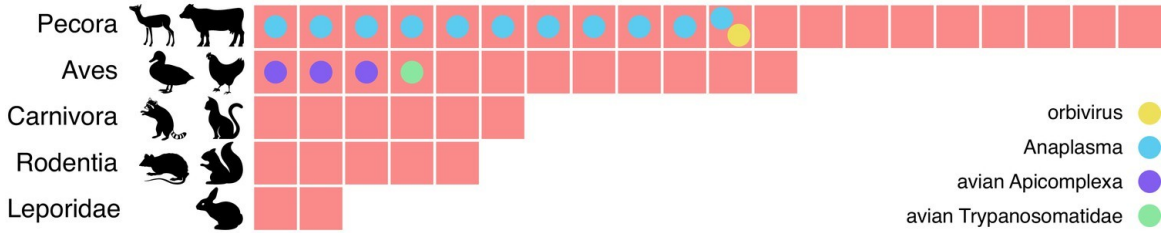
mNGS to identify vectors, emerging pathogens, and reservoirs

mNGS on 148 wild mosquitoes reveals:

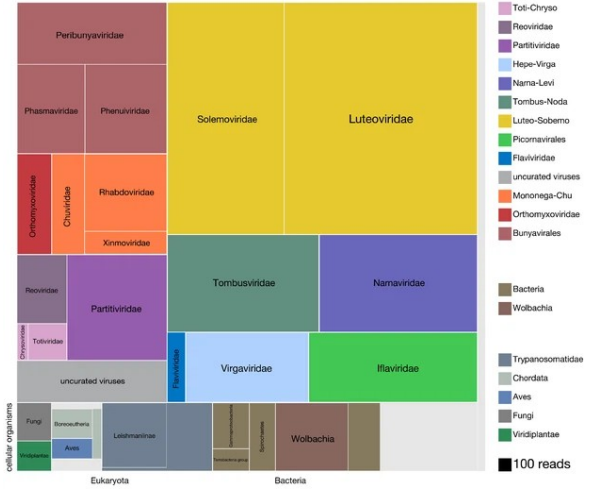
Mosquito speciation by geography



Sources of blood meals in individual mosquitoes



Microbial composition of mosquitoes



Use case: Batson *et al.*, eLife 2021

mNGS to identify vectors, emerging pathogens, and reservoirs

mNGS on a single sample type gives such rich biological information!

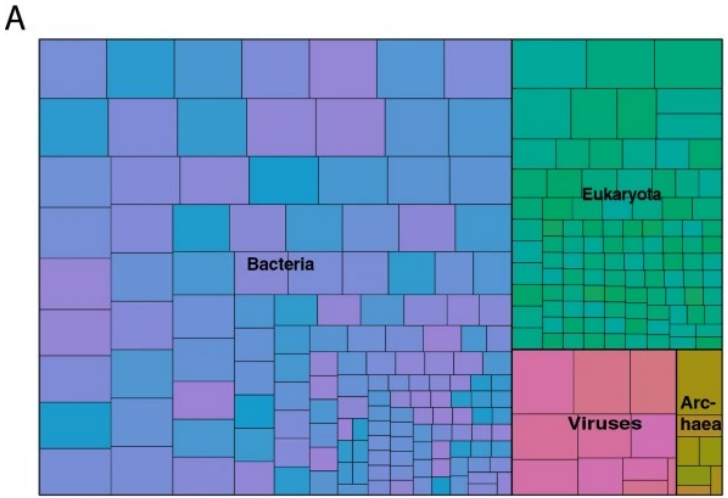
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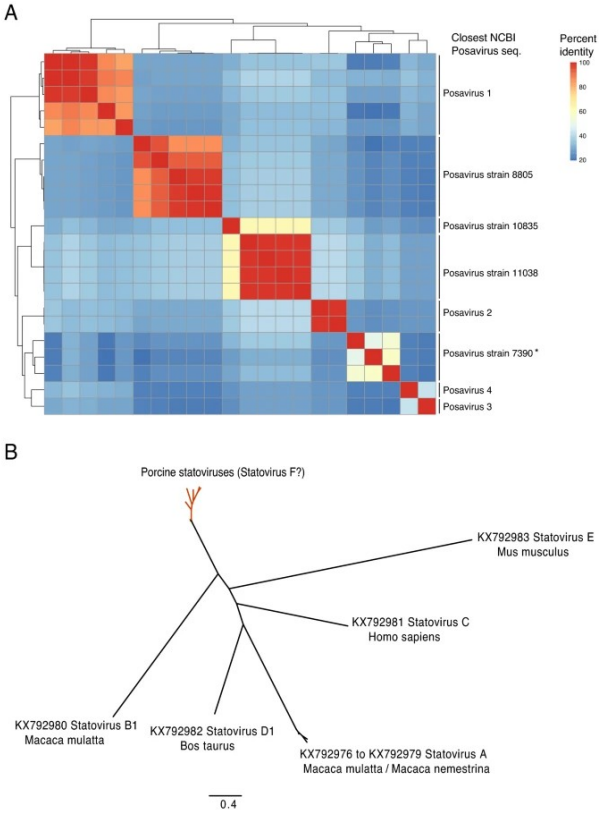
Sources of blood meals in individual mosquitoes

Use case: Ramesh *et al.*, Scientific Reports 2021

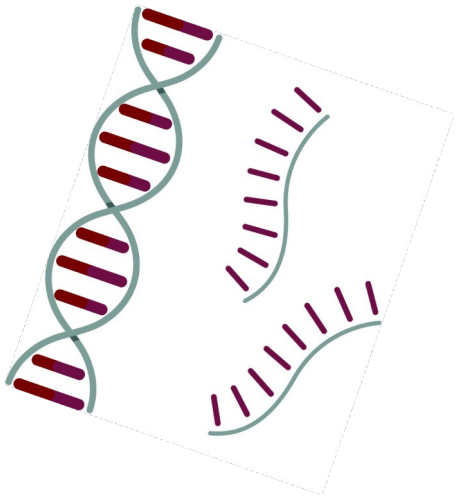
Metagenomic characterization of swine slurry in a North American swine farm operation

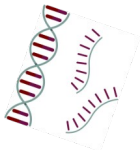


1792 viral genomes, of which 554 were novel or divergent!!!



xNA extraction





xNA extraction

Key points:

✓ Minimize cross-contamination

✓ Preserve sample quality

- Before extraction:
  - Spin down to remove host cells (extract supernatant)
- Choose type:
  - DNA:
    - Detect bacteria, parasites, fungi, DNA viruses
  - RNA:
    - All the above + RNA viruses (!)
    - Requires optimal sample quality, degrades very fast
    - Keep cold at all times!
  - TNA (total nucleic acid):
    - DNA + RNA
    - Requires DNAse I step for RNA-mNGS
- Column vs bead-based extraction

xNA extraction

Zymo DNA+RNA extraction kits



Quick-DNA/RNA Microprep Plus

50 preps for $399

Column-based



Quick-DNA/RNA Miniprep Plus

50 preps for $399

Column-based



Quick-DNA/RNA Pathogen Magbead

96 preps for $293

Bead-based

Requires extra:

- ✔ Magnet stand (CZB Engineers)
- ✔ β-mercaptoethanol

Extra reagents to buy:

- ✔ Molecular grade ethanol (200 proof) for washes
- ✔ Isopropanol to add to concentrated wash buffers
- ✔ DNase I included in regular quick-DNA/RNA kits, but not in Pathogen kits
  - Available separately from Qiagen: RNase-Free DNase Set, $512 for 250 samples

xNA extraction

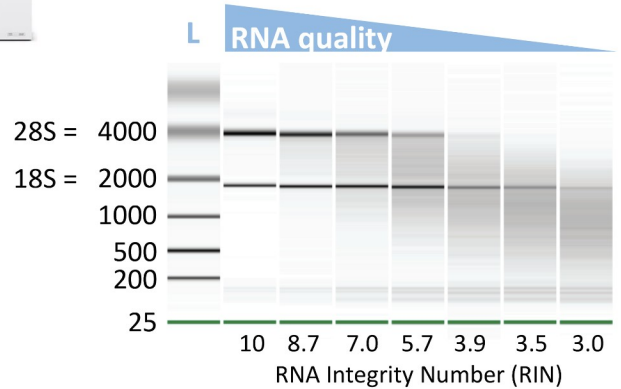
Key point:

✓ xNA quality is crucial

- Quality control:
  - Quantity (ng/µl):
    - Qubit: RNA or DNA High Sensitivity (HS) Assay Kit (ThermoFisher)
  - Quality (length):
    - Capillary electrophoresis
      - Tapestation (Agilent)
      - High Sensitivity DNA or RNA ScreenTape
    - Gel electrophoresis
      - Regular agarose gel for DNA
      - TBE-urea gel for RNA



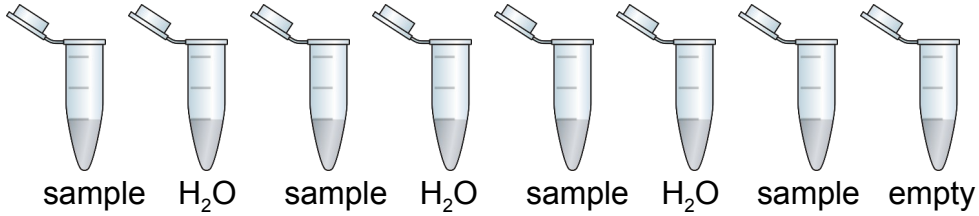




xNA extraction

Keep in mind:

- Magnetic stands for 1.5 ml tubes usually accommodate only 8 tubes at a time
- Cross-contamination should be avoided as much as possible!!
  - Process samples with high pathogen content separately from samples with low pathogen content (if you can)
  - Use water controls to estimate the level of cross-contamination
    - Space out samples with water controls, and/or empty spots in the magnetic rack
    - E.g. for 4 samples:



Next : 03_Lab setup and sample prep